MILITARY SPECIFICATION

DISPENSING SET, COUNTERMEASURES AND CHAFF
AN/ALE-39

This amendment forms a part of Military Specification MIL-D-81694(AS) dated 2 February 1972 and has been approved by the Naval Air Systems Command, Department of the Navy and is available for use by all Department and Agencies of the Department of Defense.

Previous issues of this specification and amendments are hereby declassified. Cite this amendment as authority for declassification.

PAGE 5

3.3.1: Change 2.5 pounds to 2.1 pounds.

PAGE 6

* 3.3.6: Delete in its entirety and substitute:

"Interference control. - The equipment shall operate within the limits specified by the tests listed in Table I, Class ID of MIL-STD-461A, as modified herein by Paragraph 3.3.6.2. Specific requirements of the equipment either augmenting or deleting those listed herein may be set forth in the purchase description by the procuring activity."

PAGE 6

* 3.3.6.2: Add new paragraph as follows:

"Test parameter exceptions. - The following relaxations to the limits of MIL-STD-461A and Notice 3 are allowable when the programmer and sequencer switches are interconnected and operating together with 1/2 Ohm, 10 watt resistors used as loads in place of impulse cartridges.

Test	Deviation (db)	Frequency (HZ)
CE03 and CE04	+20	20K
Broadband Emissions	+43	150K
Broadband Emissions	+25	1.5M
Broadband Emissions	+29.5	2M

FSC 5895

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Broadband Emissions	+26	25M
Broadband Emissions	+26	50M
RE02 (Broadband Emissions)	+ 8	14K
RE02 (Broadband Emissions)	+ 8	140K
RE02 (Broadband Emissions)	+ 7	500K
RE02 (Broadband Emissions)	+21	25M
RE02 (Broadband Emissions)	0	25M
RE02 (Broadband Emissions)	+ 7	35M
RE02 (Broadband Emissions)	+10	50M
RE02 (Broadband Emissions)	+15	50M
RE02 (Broadband Emissions)	+20	100M
RE02 (Broadband Emissions)	+15	100M
RE02 (Broadband Emissions)	+10	1G

Note (1) All testing will be accomplished in singles mode. Test limits do not apply during reset. Test limits are established by straight-line interpolation between points given. The above relaxations to the limits of MIL-STD-461A are in addition to the relaxations granted by Notice 3 to MIL-STD-461A.

Note (2) Shortening of Burst Intervals and Salvo Intervals during CS02 testing is permitted for interference signals of 0.4 Vrms or greater above 750 KHZ when applied to the 5 Vac lighting line. Uncontrolled firings are not permitted during any test."

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* 3.3.10.2.1: Delete in its entirety and substitute:

"The programmer shall operate continuously at a maximum ambient temperature of +55°C with the lighting circuit energized and +95°C with the lighting circuit not energized. For the purpose of MIL-E-5400, short term, intermittent, or high temperature storage extremes do not apply."

PAGE 8

* 3.3.12.2: Delete in its entirety and substitute:

"Input power for lighting of the programmer shall require not more than 10 watts at 5.0 volts, AC, 400 HZ. No lighting power is required for the dispenser or sequencer switch. No lighting power is required for the programmer above +55°C ambient."

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3.3.12.4: Change 900 mw to 1.5 watts.

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3.5.2.2..1: Delete "14/28 VAC".

PAGE 10

3.5.2.3: Change "2.5" to "2.1" pounds.

PAGE 10

3.5.2.5.1: Change "Automatic" to "Program".

PAGE 10

3.5.2.5.1.1 (2): In second sentence add "0.50" to intervals listed.

PAGE 11

Table I: Change "Automatic" to "Program".

PAGE 12

Table I, Note 1C: Delete Note 1C in its entirety and substitute:

"Burst Quantity (R) and Burst Interval (Fixed No.) - Four, five, or six bursts at the fixed time interval. The number bursts shall have equal probability of occurrence. Any additional salvos will occur at the salvo interval."

PAGE 13

3.5.2.5.2: Change first sentence to read:

"Single Mode. The programmer shall provide for immediate (within 0.125 second) dispensing of a single desired payload each time the single mode is selected on the cockpit control unit and the press-to-dispense switch is pushed in."

PAGE 14

3.5.2.5.5: Change "Flare-Rapid Fire sequence" to read "SALVO Flares sequence."

PAGE 14

3.5.2.6: Add to end of sentence "(or equivalent)."

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PAGE 14

3.5.3.1: In 2 places change "squib" to "impulse cartridge". This is the correct designation per NAVAIR 11-100-1.

PAGE 15

3.5.3.1.1: In 6 places change "squib" to "impulse cartridge".

PAGE 15

3.5.3.1.2: In one place change "squib" to "impulse cartridge".

PAGE 22

4.5.1: Change "voltate" to "voltage".

PAGE 26

Figure 1: Change pin letters on programmer:

PIN	NEW	FUNCTION
A	G	Man. Chaff
B	H	Mult.
C	J	2^1
D	U	2^0
E	L	Chaff ctr.
F	M	2^2
G	B	+28VDC
H	C	L-20
J	E	Clk INH
K	R	Ext. clk
L	D	GND
M	F	L-10
N	N	Spare
P	T	Flare ctr.
R	K	5Vac
S	A	Reset
T	P	R-10
U	S	Jam ctr.
V	V	R-20

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Figure 2: Change following dimensions:

From	To
8.440	8.60
6.250	6.70

Change: Note 1 from 1.9 to 2.1 lbs

Note 2 from 0.9 to 1.5 watts

Figure 5: Under Note 1, add decimal point to read weight

Housing 4.66 lbs.

Dispenser (loaded) 21.30 lbs.

* Figure 6: Delete in its entirety and substitute new Figure 6.

30.1.5.1: Delete in its entirety and substitute:

"Power Control. A switch position shall be provided on the Payload Selector Control (see 30.1.5.2) rotary switch to control the primary power to the AN/ALE-39 dispenser system. This position shall be labeled "OFF". In the "OFF" positiion primary power shall not be provided to the programmer and control units. In any other position (SALVO F, C, F/S, F/M, J) primary power shall be provided to the control and programmer units of the AN/ALE-39 dispensing system."

30.1.5.2: Delete in its entirety and substitute:

"Payload Selector Control. A single six-position rotary switch shall be provided for selection of type payload to be dispensed. The switch positions shall have the following functions and labels in a clockwise rotation: - Salvo flares ((SALVO F (PULL)), poweroff (OFF), Chaff (C), Flare-single (F/S), Flare-multiple (F/M), and Jammer (J)). See subparagraph 30.1.5.2.1 below regarding salvo flares control configuration. A red press to-dispense momentary switch, labeled "DISP", shall be located in the center of the Payload Selector Control to initiate the dispensing sequence. It shall operate in conjunction with the Payload Selector Control and the Mode Select switch (30.1.5.3)."

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30.1.5.2.1: Delete in its entirety and substitute:

"Salvo Flares Control. A guarded pull-to-turn position on the rotary payload selector control switch shall be provided to actuate the programmer flare-firing sequence (reference paragraph 3.5.2.5.5). The switch position shall be labeled as indicated in paragraph 30.1.5.2 above, and it shall have a striped background as specified in MIL-C-81774, paragraph 3.5.5.6. When the salvo flares position is selected it shall initiate the flares salvo without having to press the "press-to-dispense" switch."

PAGE 34

30.1.5.2.3: Change "Reference 3.5.5.4)." to read "(reference 3.5.2.5.4)."

PAGE 35

Change "30.1.5.2.3" to read "30.1.5.3."

PAGE 35

30.1.5.3 Mode Control: Delete in its entirety and substitute:

"Mode Select. A single, long throw, positive action, three-position toggle switch shall be provided to operate in conjunction with the payload selector control and press to-dispense button to select the dispensing mode. The switch shall be labeled Single (SNGL), Standby (STBY), and Program (PRGM)."

"SNGL" When the switch is moved to "SNGL" or lower position and the press-to-dispense button is pressed a single burst of the respective payload shall be dispensed within 0.075 seconds. The single mode may also be initiated while on automatic dispensing sequence is in operation by selecting "SNGL" and pressing the press-to-dispense button."

"PRGM" When the switch is moved to the "PRGM" or upper position and the press-to-dispense button is pressed, the programmed sequence command for the selected payload (reference 30.1.5.2) shall be entered into the programmer. The programmer shall control the rates and quantities of the appropriate payload until the dispensing program is completed. Once a programmed dispensing sequence has been initiated it must be completed before another programmed dispensing sequence can be initiated. A programed sequence may be interrupted by a manually initiated single burst, after which the programed sequence will then resume."

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30.1.5.4: Change "(Pin A)" to "(Pin G)". Add the following sentence - "This switch shall function independent of the payload selected (C, F/S, F/M, J) on the Payload Selector Control."

PAGE 36

30.1.5.5 Warning Receiver Override. Delete in its entirety and substitute:

"Warning Receiver Override. A two position toggle switch shall be provided to disconnect the interface circuit in the control unit from the automatic AN/ALR-50 warning receiver (J2-M) input. The switch shall be labeled "WR AUTO CHAFF" in the upper position and "OFF" in the lower position. In the CHAFF-AUTO fire command code when initiated by the warning receiver. A repeat delay circuit shall be utilized to prevent the CHAFF-AUTO code from being generated more often than 30 seconds apart."

PAGE 36

30.1.5.6, after first sentence: Delete and substitute:

"The logic interface between control and programmer units is positive-true with logical "0" defined as 0.1 to +1.0 volts, and logical "1" defined as +10.0 to +12.0 volts. In the logical "0" conditions, the control unit circuitry shall be capable of "sinking" 3.2ma from the programmer unit. The control unit shall provide an open circuit to the programmer to satisfy the logical "1" condition."

PAGE 36

30.1.5.6.1: Change to as follows:

1. "Chaff Manual" to "Chaff Single"
2. "Chaff-Auto" to "Chaff Programmed"
3. "Flare Single-Manual" to "Flare Single"
4. "Flare Single-Auto" to "Flare Single Programmed"
5. "Flare Multiple-Manual" to "Flare Multiple"
6. "Flare Multiple-Auto" to "Flare Multiple Programmed"
7. "Jammer Manual" to "Jammer Single"
8. "Jammer Auto" to "Jammer Programmed"

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30.1.5.6.1 Fire Command Encoder: Change function of item 9 from "Rapid Fire-Flare" to "SLAVO Flares".

PAGE 37

30.2.1, in one place change "squibs" to "impulse cartridges".

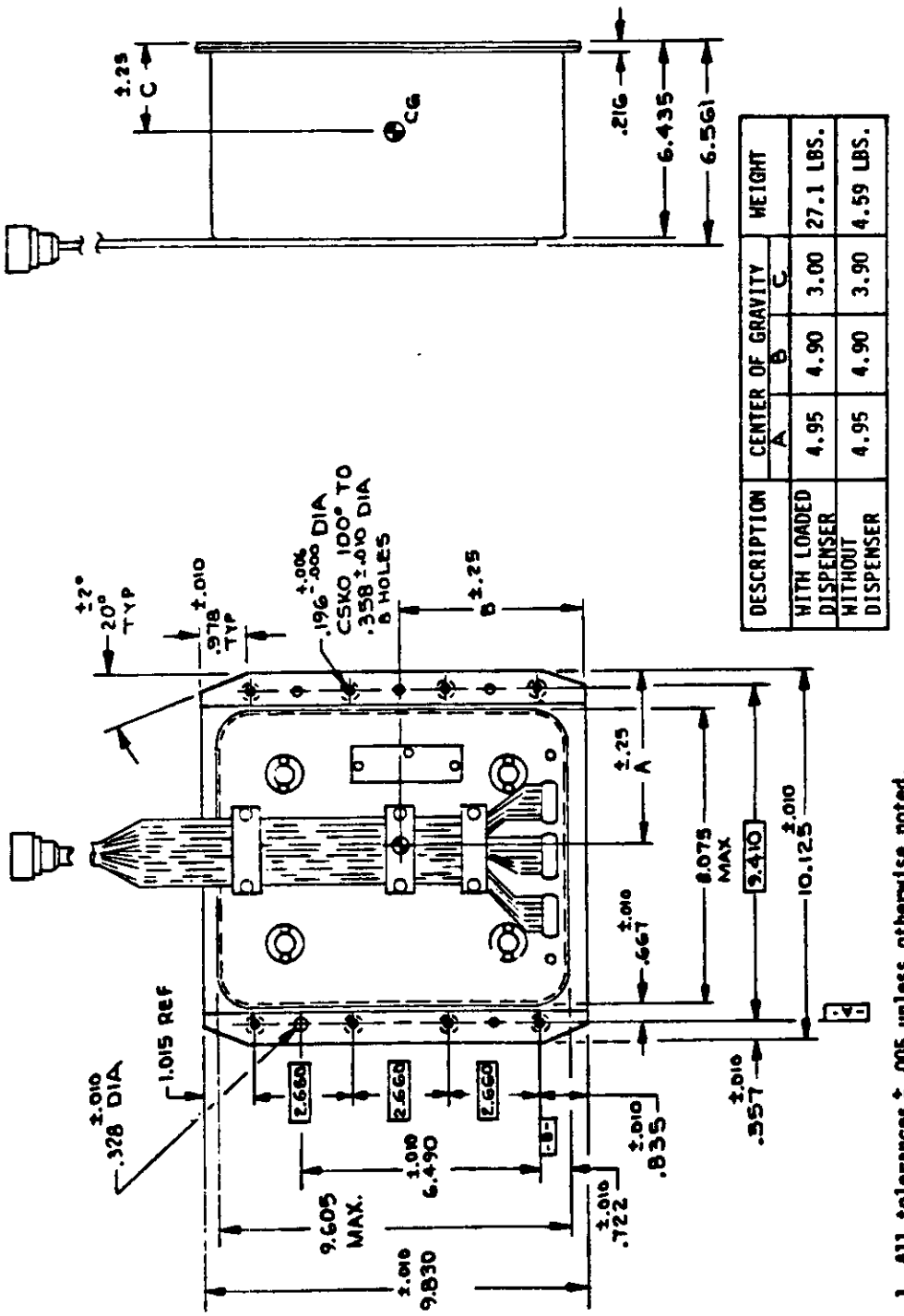
PAGE 38

30.3.1, in 2 places change "Squib" to "impulse cartridges".

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DESCRIPTION	CENTER OF GRAVITY			WEIGHT
	A	B	C	
WITH LOADED DISPENSER	4.95	4.90	3.00	27.1 LBS.
WITHOUT DISPENSER	4.95	4.90	3.90	4.59 LBS.

1. All tolerances ±.005 unless otherwise noted.
2. This drawing is not to scale.

Figure 6 - MX-7721/ALE-29A Dispenser Housing

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